

Course Outline & Introduction

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Multivariate Statistics for Applied Data Science (ADSC 4050)

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THOMPSON RIVERS UNIVERSITY

Topics

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Course Basics

- Course: Multivariate Statistics for Applied Data Science (ADSC 4050)
- Lecturer: Ing. Sean Hellingman, Ph.D.
- Meeting times:
 - Monday 11:30 - 12:45 OM 1360
 - Tuesday 14:30 - 15:45 OM 1360

Course Educational Outcomes

1. Apply and modify existing methods of multivariate data analysis.
2. Learn and select appropriate distance metrics.
3. Select and create appropriate dimension reduction approaches.
4. Select an appropriate method of analysis given the analytic question.

Tentative Topics

- Distance metrics
- Principal Component Analysis (PCA)
- Canonical Correlation Analysis (CCA)
- Singular Value Decomposition (SVD)
- Factor Analysis (FA)
- K-means clustering
- Hierarchical clustering
- Discriminant analysis
- Classification trees
- Random forests
- Boosting
- Bagging
- Nearest Neighbours
- Resampling methods

Software

- R within the R Studio environment.
 - *Python will be accepted for some topics.*

Moodle

- All important course information will be located on the course moodle page.
 - Course information and announcements
 - Lecture slides
 - Examples
 - Assignments
 - Grades

Lectures

- Content is focused on the applied side of statistical theory.
- Combination of lecturing and working through examples.
- Please ask questions as they arise!
- Practice *exercises* are included at the end of the lecture slides for your convenience.
 - Further examples will be presented in the seminars.
- **Expectations:**
 - Please be respectful to everyone.
 - **Please don't be disruptive to your colleagues.**
 - Please leave your mobile devices on silent.

Lecture Participation

- We are in a computer lab & R is loaded on the computers.
- I will post example code **prior** to lectures.
- It is strongly encouraged to work along through the examples.

Course Evaluation Weights

- Assignments 30%
- Tests 30%
- Project 40%
 - 30% Written report
 - 10% Oral presentation

TRU Letter Grades

Letter Grade	Numerical Grade	Grade Points	Letter Grade Definitions
A+	90–100	4.33	Excellent. Superior performance showing comprehensive, in-depth understanding of subject matter. Demonstrates initiative and fluency of expression.
A	85–89	4.00	
A-	80–84	3.67	
B+	77–79	3.33	Very good. Clearly above average performance with knowledge of principles and facts generally complete and with no serious deficiencies.
B	73–76	3.00	
B-	70–72	2.67	
C+	65–69	2.33	Satisfactory. Basic understanding with knowledge of principles and facts at least adequate to communicate intelligently in the discipline.
C	60–64	2.00	
C-	55–59	1.67	Pass. Some understanding of principles and facts but with definite deficiencies.
D	50–54	1.00	Minimal pass. A passing grade indicating marginal performance. Student not likely to succeed in subsequent courses in the subject.
F	0–49	0.00	Unsatisfactory. Fail. Knowledge of principles and facts is fragmentary.

Assignments (30%)

- 5 total assignments:
 - Focus on practical applications based on the lectures.
 - Assignments will be posted on moodle with due dates clearly specified.
 - Due dates can be found in the tentative calendar & on moodle.
 - The **lowest** grade will be dropped.
 - Collaboration is **encouraged** but please **submit separate and different documents**.
 - **Note:** *Due dates are not always consistent.*

Assignment Expectations (30%)

● Expectations:

- Assignments are mandatory in this course.
- Assignments will be submitted (unless otherwise instructed) as PDF files in moodle.
- Please include your name and TRU ID on your assignment.
- ***Late assignments will not be graded.***
- *LLMs can help when you are stuck, they are not to be used to complete assignments.*
 - *The student may be penalized for any solutions that appear to be directly produced by generative AI.*

Tests (30%)

- 2 total tests:
 - Test I is tentatively scheduled for **10 March during the lecture period.**
 - Test II will take place during the exam period.
 - More details about the tests will be uploaded closer to the test dates.
- **Expectations:**
 - Please leave personal items, including mobile phones, in your bag at the front of the class.
 - Please bring and be ready to present English/French language identification (TRU student card) to the tests.
 - **The TRU Academic Integrity policy is always active.**

Project (40%)

- 1 term project:
 - Designed to apply statistical methods to real-life data.
 - Groups of 1-3 students may work on the same project.
 - Tentatively due the last week of classes.
 - Presentations will be scheduled for the last week of classes.
 - *More detailed expectations will be posed soon.*

Office Hours

- Room: CT412
- Times:
 - Tuesday 16:00 - 16:50
 - Wednesday 15:45 - 16:35
 - *Or by appointment*

Email

- Email questions to: shellingman@tru.ca
- I will be happy to help when I can!
- **Email Expectations:**
 - Please, include an informative subject.
 - Please, clearly state your exact problem and what you have already tried.
 - Please do not email questions last minute (you may not get help in time).

Suggested Textbooks

- The lectures, exercises, and assignments should be sufficient.
- I will also post links to open-source resources throughout the course.
- If you want more information on these topics:
 - Roger D. Peng. *Exploratory Data Analysis with R*
 - Tiffany Timbers, Trevor Campbell, and Melissa Lee. *Data science: A first introduction*. Chapman and Hall/CRC
 - Jonathan D. Bakker. *Applied Multivariate Statistics in R*

Student Resources at TRU

- Academic Supports
- Health and Wellness
- Diversity & Equity
- Career & Experiential Learning
- Security
- Students' Union

TRU SAFE App

- App: <https://www.tru.ca/risk-management-services/security/tru-safe-app.html>
- Apple: <https://itunes.apple.com/app/id1151547903>
- Android:
<https://play.google.com/store/apps/details?id=com.cutcom.apparmor.tru>

Accessibility

- All TRU students who require accommodations are encouraged to register with Accessibility Services upon registering with TRU.
- Help determine how test and exam accommodations can be arranged.

LLM AI (ChatGPT)

- *“It works best for tasks we could do ourselves but shouldn’t waste time on, yet it can actively harm our learning when we use it to skip necessary struggles”* - Ethan Mollick
- **The instructor reserves the right to question and penalize the student on any solutions that appear to be directly produced by generative AI.**

Course Outline

- Please refer to the official Course Outline document for additional information about the course.

Exercise 1

- Take some time to review ADSC2020:
 - How do we assess the accuracy of our forecasts and predictions?
 - How did our models perform when there were many explanatory variables?
 - Do you think there are any weaknesses with the methods that we have discussed?